



Social Transformation Fellowship: Skin of Color Catered Machine Learning Model

Pranav Desurkar, Aarti Sridhar, Atishay Kumar, Darrin Zhou

Abstract

- We found a problematic discrepancy in the diagnosis Skin of Color (SoC) and non-SoC in dermatological settings, which was a need that other projects were not focusing on.
- Our team is constructing a ML-powered diagnostic tool that can classify skin conditions frequently misdiagnosed on SoC, currently starting with eczema.
- Our model surpassed an industry competitor on relevant performance metrics for image classification by dataset augmentation.
- The range we used to classify skin tones is known at the **Fitzpatrick scale** with tones ranging from F1-F7 with F1 being the fairest and F7 being the deepest.

Problem and Context

- ML-powered dermatology tools to reduce diagnostic errors and improve access to care. Current tools often replicate existing disparities in healthcare, especially when diagnosing individuals with Skin of Color (SOC).
- Dermatological disparities in the U.S. are caused by the underrepresentation of SoC in medical training resources. According to Narla et al. (2022), "47% of dermatologists reported feeling inadequately trained to diagnose conditions on skin of color".

Stakeholder Research

- Our team interviewed medical students, a nursing student and healthcare professionals in order to gain insight on:
 - Disparities in SoC Healthcare education
 - Sentiments surrounding artificial intelligence in healthcare
- Our stakeholders said:

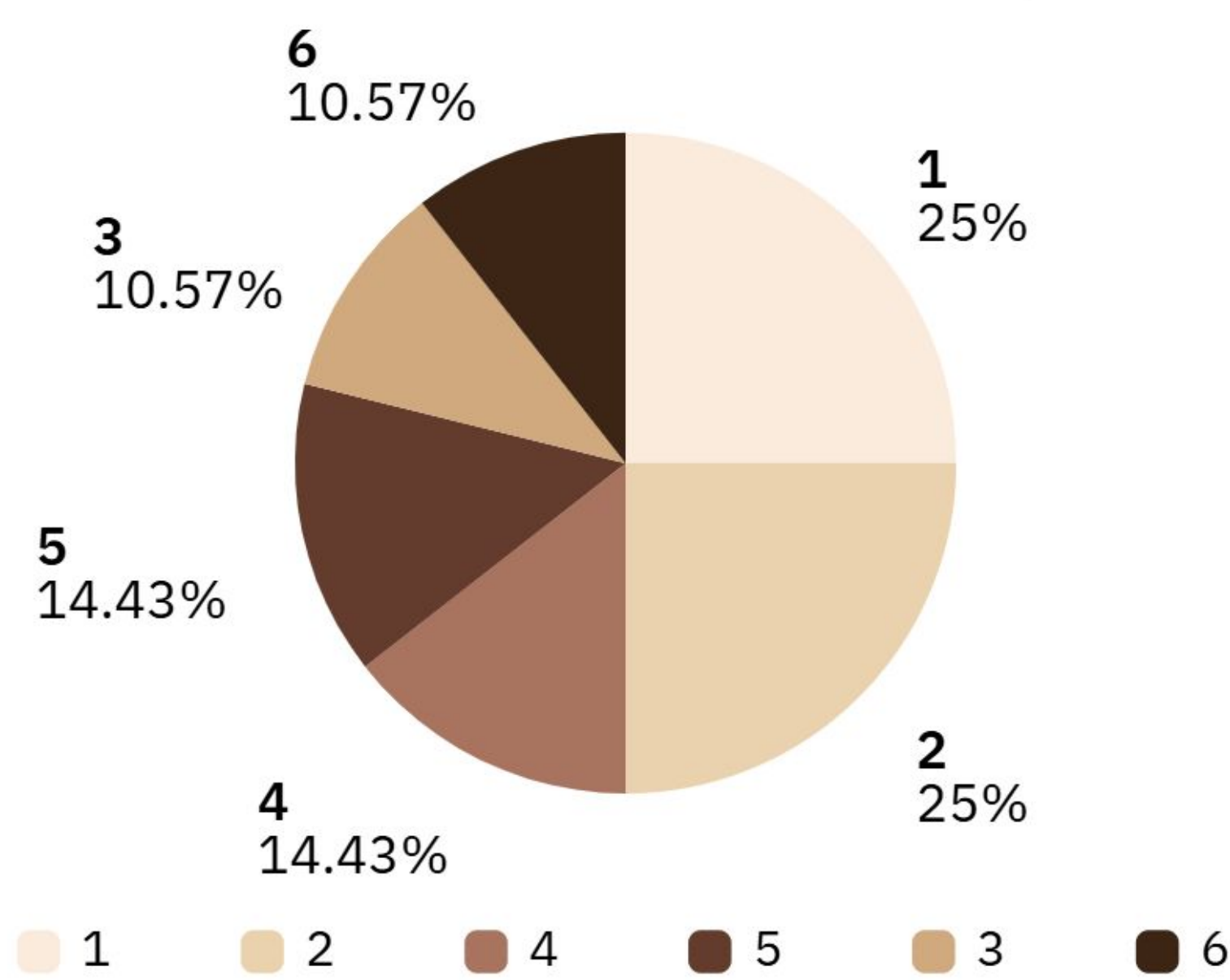
“ML classification tools would be useful in healthcare spaces that don’t have specialists”

“... not much in-depth content on what wounds and other conditions look like on darker skin was available”

Our Implementation

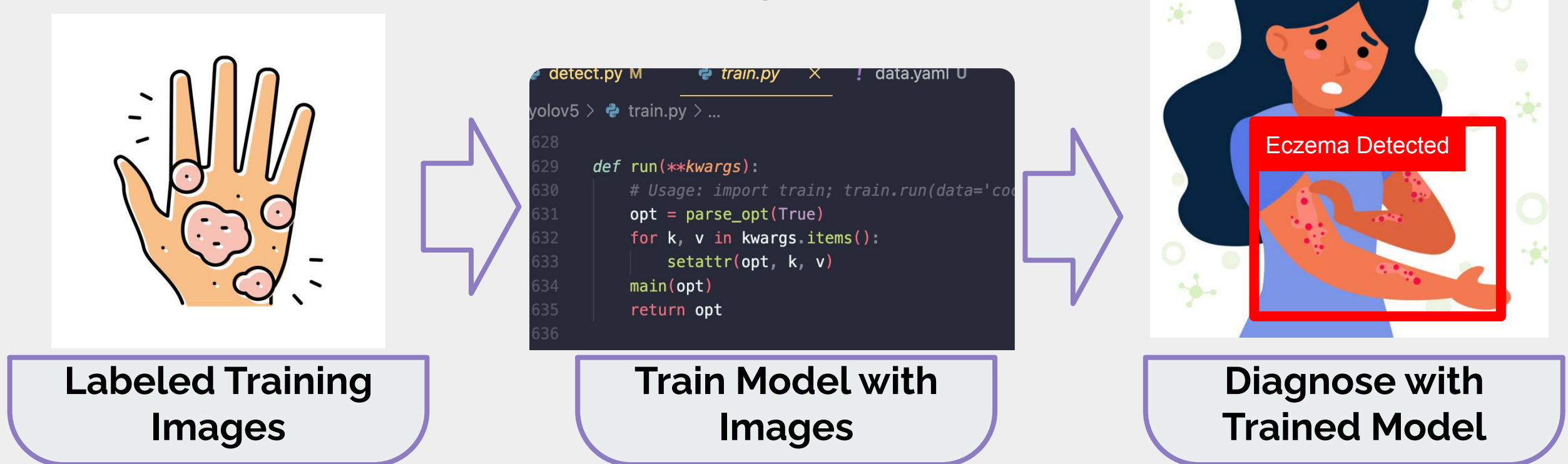
Step 1: Assembling a Representative Dataset

Distribution of skin tones across SoC Dataset in terms of Fitzpatrick Scale Rating



Our team ensured that our training and testing dataset consisted of 50% images of SoC which is represented by the Fitzpatrick scale ratings 3-7 for both images featuring eczema as well as images containing other similar conditions such as bacterial skin infections.

Step 2: Training our Model



Then we trained a standard ResNet model on the augmented dataset. In this process our model imitates a human brain, rapidly learning underlying patterns that are key distinguishing between eczema and other conditions.

Step 3: Evaluate against competitor

Finally, we evaluated the model against a testing set that featured images on a range of Fitzpatrick skin types **F1-F7**. We then compared our model's performance to another commonly used dermatological model built on ResNet.

Acknowledgements

We would like to thank the Barger Leadership Institute and advisor Fatema Haque for providing our team with the guidance and resources necessary to conduct this community focused project. In addition, we would like to thank Dr. Blair Jenkins, Dr.Trilokraj Tejasvi, and nursing student Xochitl Trevino-Kwasnik for providing our team with guidance and insight.

References

1. Narla, S., Heath, C. R., Alexis, A., & Silverberg, J. I. (2023). Racial disparities in dermatology. *Archives of dermatological research*, 315(5), 1215–1223. <https://doi.org/10.1007/s00403-022-02507-z>
2. Adelekun, A., Onyekaba, G., & Lipoff, J. B. (2021). Skin color in dermatology textbooks: An updated evaluation and analysis. *Journal of the American Academy of Dermatology*, 84(1), 194–196. <https://doi.org/10.1016/j.jaad.2020.04.084>

Results

Performance of SoC catered Training Model vs Standard Dermatological model

Metric	SoC-catered model			Standard Dermatological Classification model		
	All Skin (F1-F7)	SoC skin (F3-F7)	non-SoC skin (F1-F2)	All Skin (F1-F7)	SoC skin (F3-F7)	non-SoC skin (F1-F2)
Accuracy	0.83	0.85	0.82	0.71	0.67	0.74
Precision	0.83	0.77	0.85	0.96	0.29	0.99
F1 Score	0.90	0.77	0.86	0.61	0.04	0.75
Auroc Score	0.90	0.92	0.88	0.72	0.50	0.79

Relevance of Metrics in Context of Classifying Eczema

Metric	Meaning
Accuracy	Correct predictions / total predictions
Precision	Correct eczema predictions / all eczema predicted; our model shows increased precision
F1 Score	How well model balances missing vs. false alarms
AUROC Score	Ability to distinguish eczema vs. non-eczema

Discussion

- **Outcome** - Our representative dataset augmentation, as dictated by the Fitzpatrick scale, led to higher performance in classification of eczema on SoC.
- **Problems** - Our biggest hurdle was finding substantial, representative datasets, especially in classification of other, rare conditions. Getting clearance for data from hospitals also proved to be difficult and slow. We've developed our challenge resilience habits, which helped us to move forward.
- **Next Steps** - We'll experiment with dataset augmentation and other bias-reduction techniques to further optimize the model's performance. We plan to develop a web application for providing reliable diagnosis for multiple conditions.

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Performance of SoC catered Training Model vs Standard Dermatological model						
Metric	SoC catered model			Standard Dermatological Classification model		
	Overall Performance	SoC skin (F3-F7)	non-SOC skin (F1-F2)	Overall Performance	SoC skin (F3-F7)	non-SOC skin (F1-F2)
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Abstract

- Because Machine Learning Models often pick up on the human biases, we wanted to see if there was a way that a learning model could bridge the gap between its diagnosis rate on Fitzpatrick skin types IV-VI (the darker range) and the lighter range of skin tones given by Fitzpatrick skin types I-III
- To accomplish this goal, we trained a Machine Learning Model on a dataset that had greater representation of darker skin tones, and compared the training of our model to other published models

Stakeholder Research

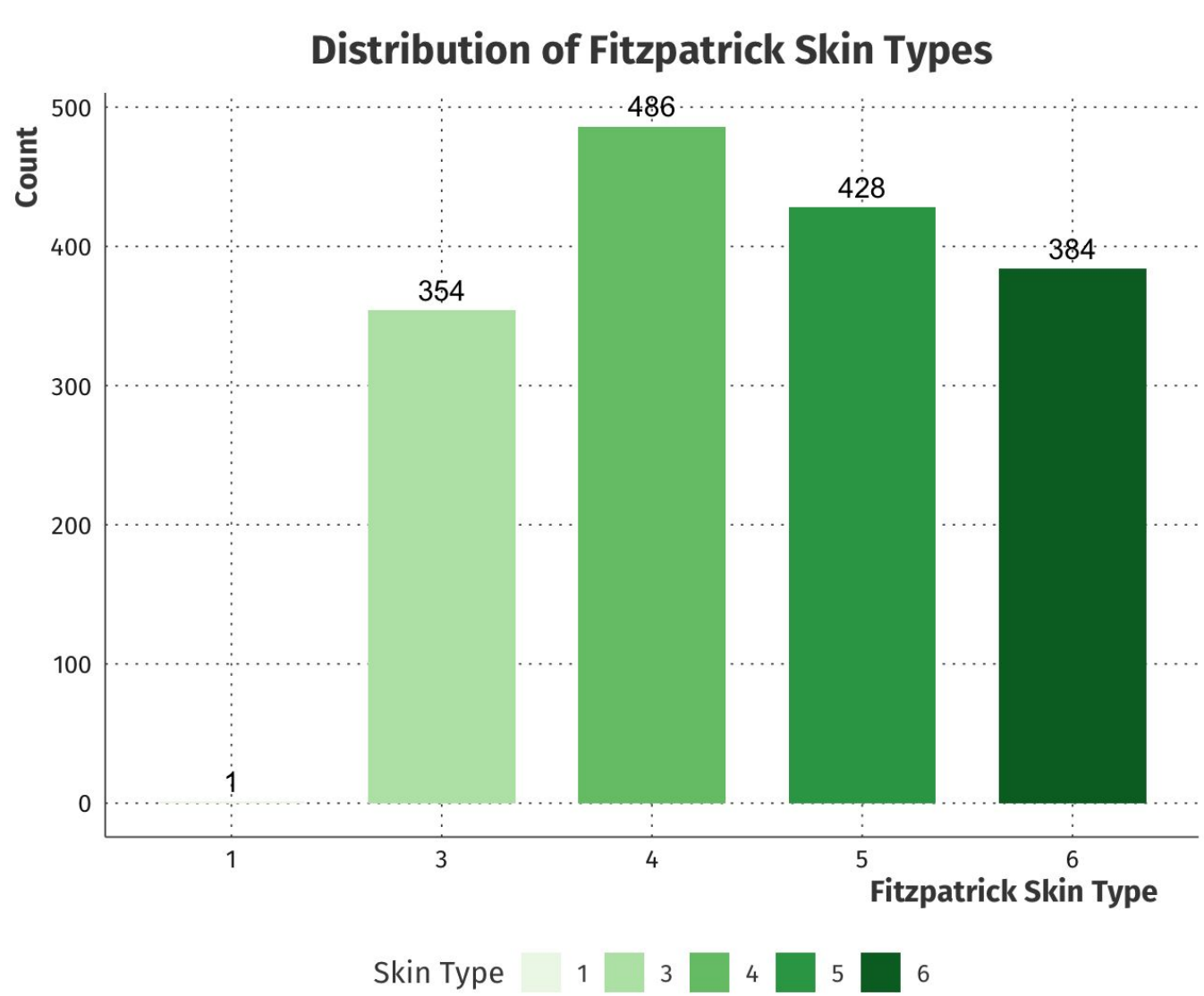
Results

Discussion (Lesson Learned)

Background/Need

- There is a need for more equitable machine learning models in Dermatology as there are discrepancies between accuracy in diagnosis rate for many ethnic groups in the United States
- According to Narla et. al., in 2022, 47% of dermatologists felt that their training was inadequate for making diagnoses on skin of color
- In another study by Adelekun et. al., in 2021, they found various racial biases in Dermatology textbooks, including underrepresentation of conditions that disproportionately affect racial groups with darker skin tones and images of darker skin tones were more often excluded in representation of common conditions like acne

Implementation Plan



Next Steps

Acknowledgements

Thank you to the Barger Leadership Institute And special thanks to Fatema Haque for providing our team with guidance on how to structure a community focused project.

In addition, we would like to thank Dr. Blair Jenkins and Dr.Trilokraj Tejasvi for providing our team with guidance on the direction of our project.

References

1. Narla, S., Heath, C. R., Alexis, A., & Silverberg, J. I. (2023). Racial disparities in dermatology. *Archives of dermatological research*, 315(5), 1215–1223. <https://doi.org/10.1007/s00403-022-02507-z>
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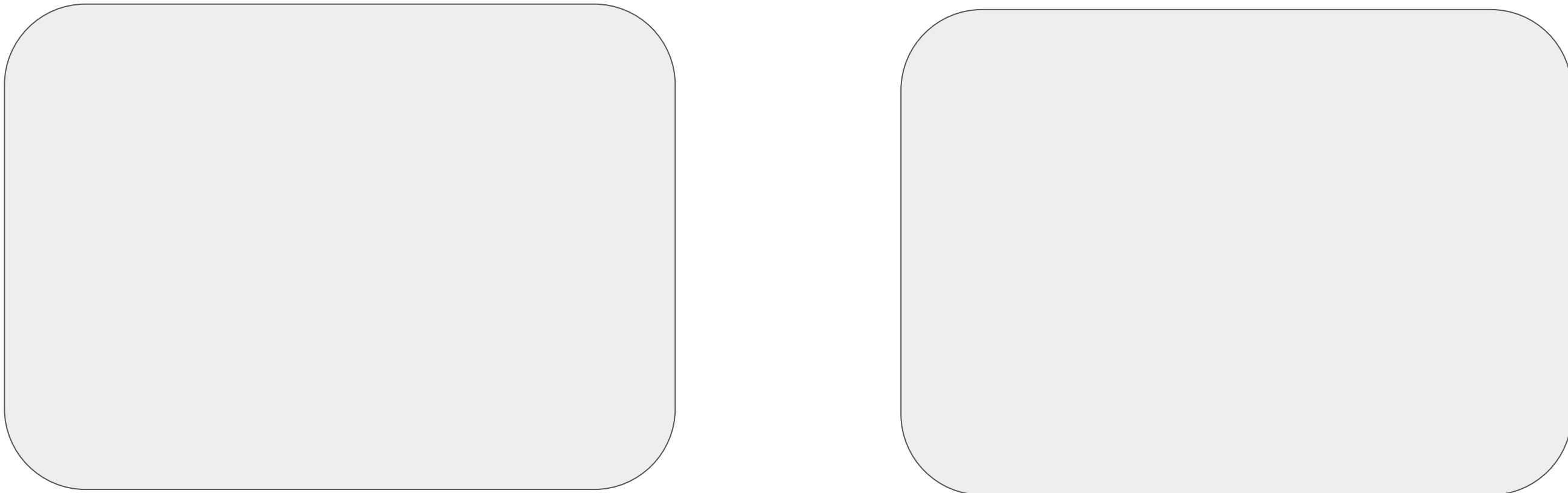
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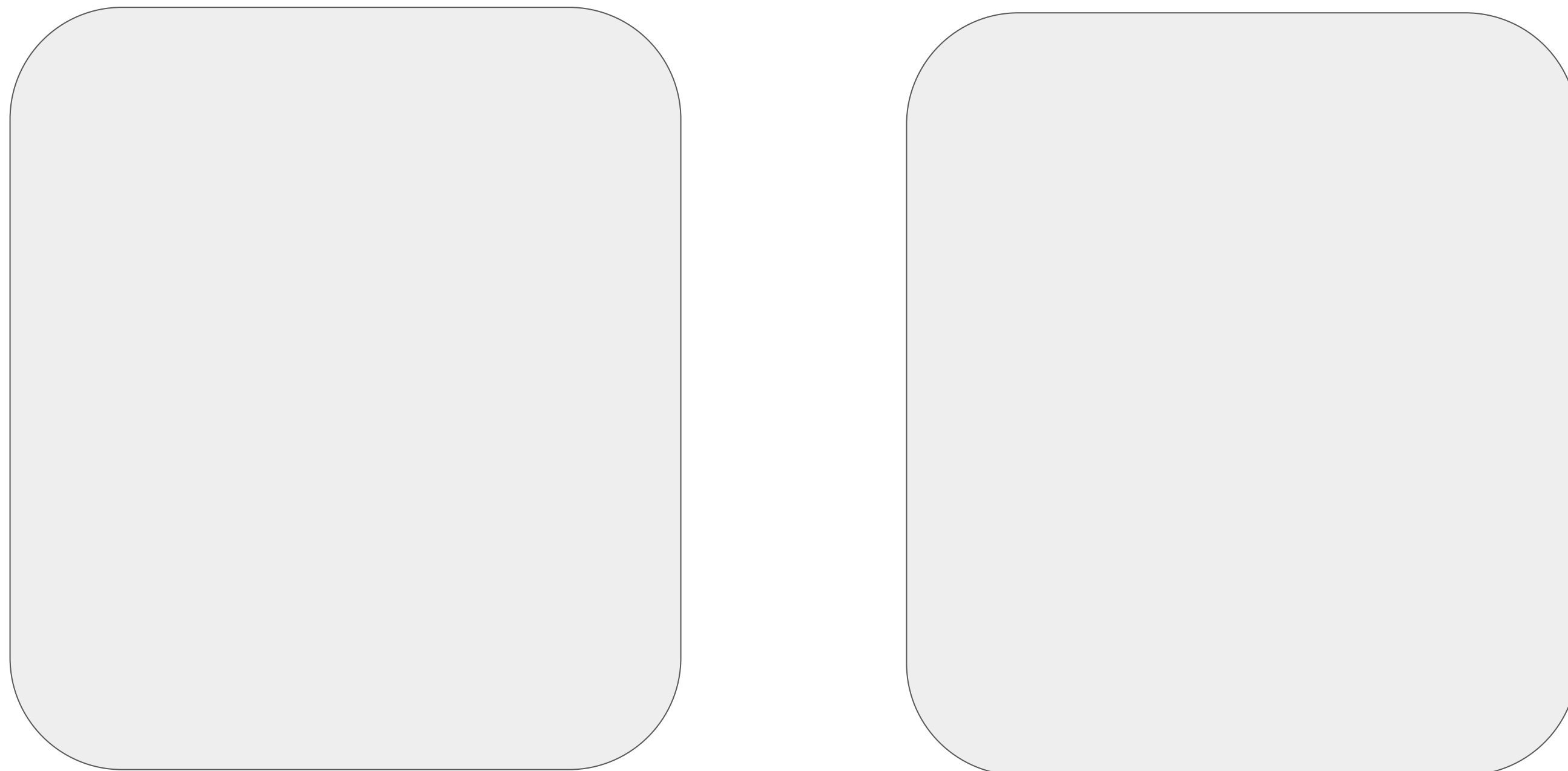
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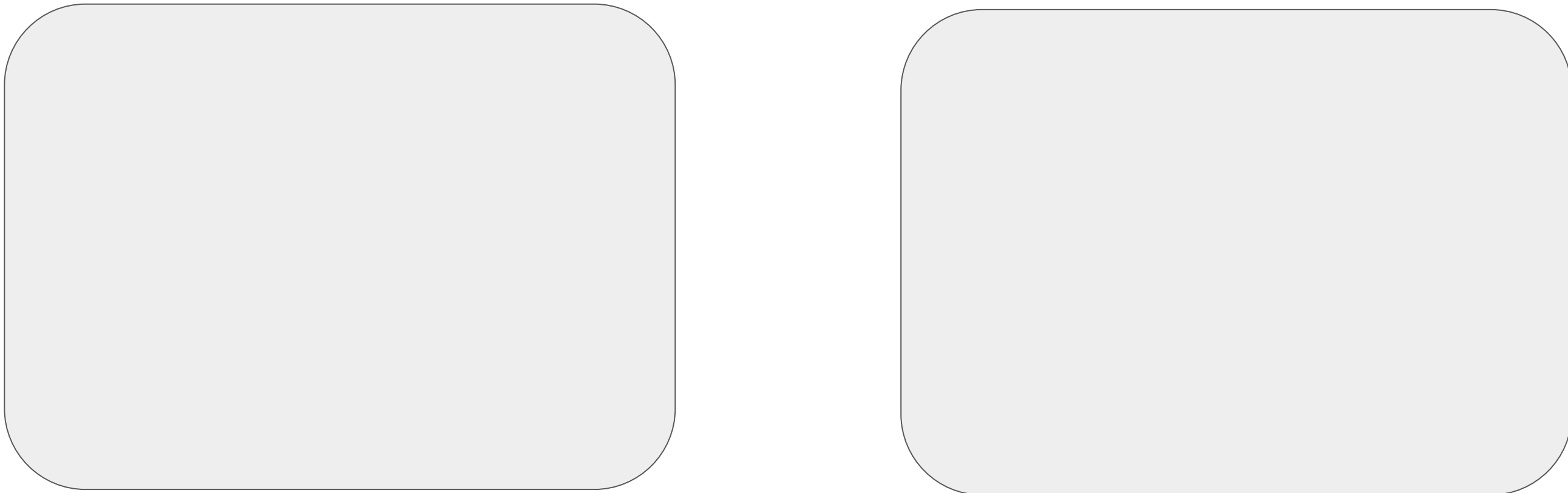
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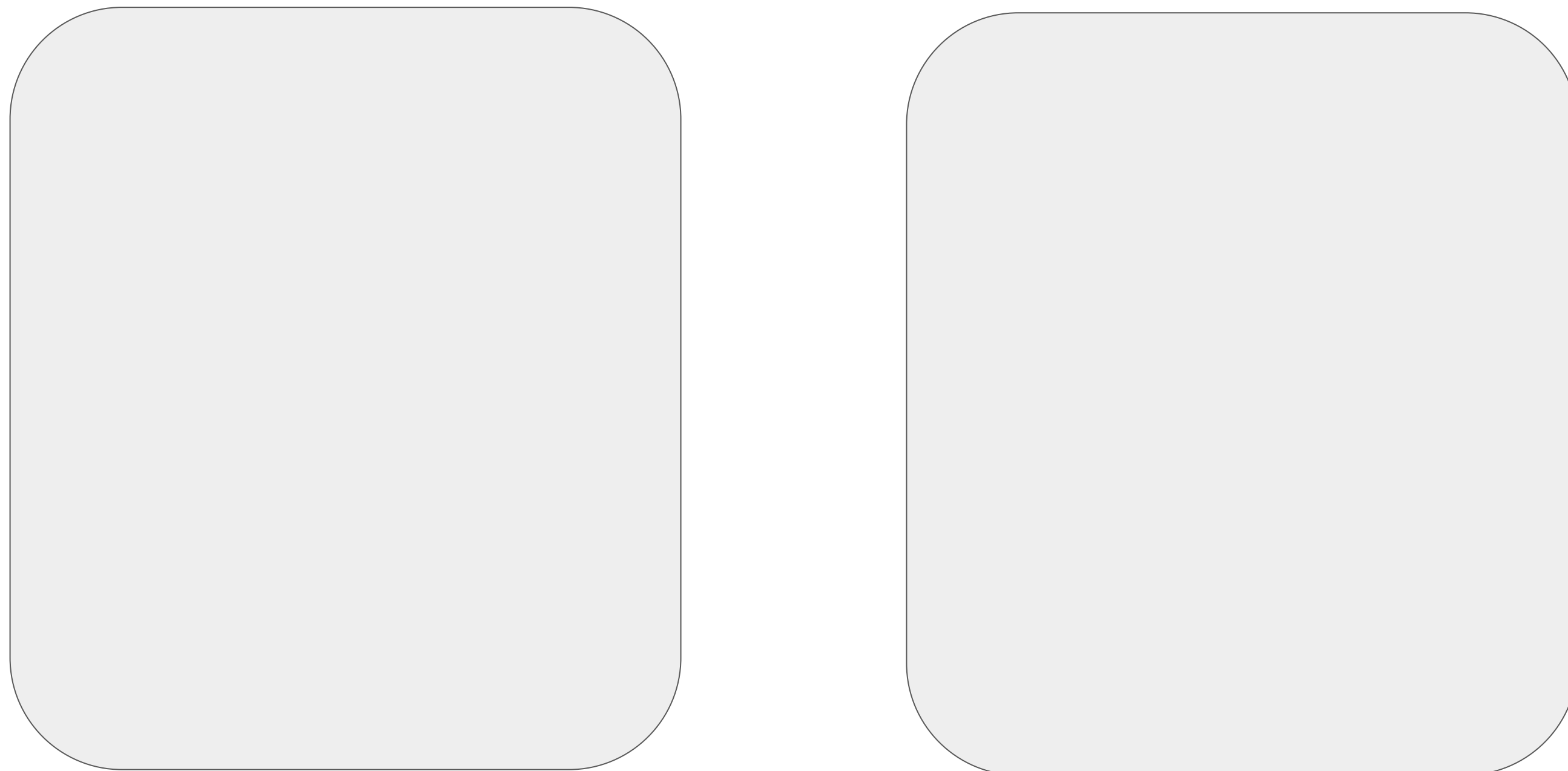
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accounting for morphological variation in the adult population

Introduction

- 7.3 million motor vehicle accidents occur annually in the U.S .
- Finite element(FE) head models play a critical role in a variety of safety applications, serving purposes of safety equipment design, such as injury risk assessment and the design of protective helmets.
- Current models are based on a mid-sized male which doesn't account for the size and shape variability of morphological dimensions across a broader adult population.(Hu et al., 2021).
- Reports from the IIHS indicate that women are more likely to be injured than men in a car crash.

Objectives

- Develop a statistical head geometry model that accounts for size and shape variations among the entire adult population, including the elderly population
- Enhance efforts in developing a statistical head geometry model that represents variations in age, sex, stature, BMI, head length, head breadth, and trigion-to-top of head distance.

References

- Bureau of Transportation Statistics. (2021). Motor Vehicle Safety Data.
- Hu, J. & Jones, M. (2021). Adaptive Design for Human Mobility.

Results

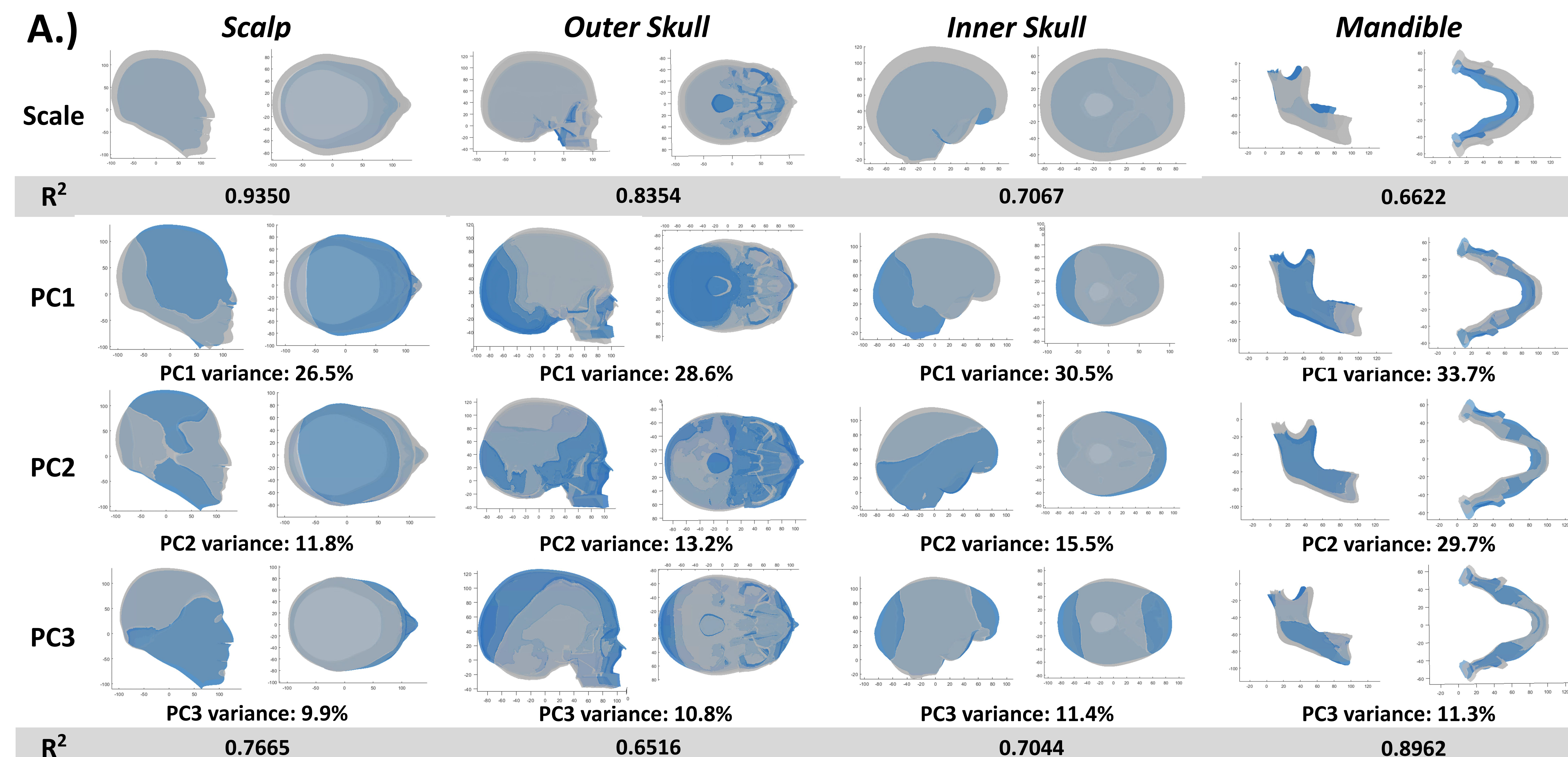
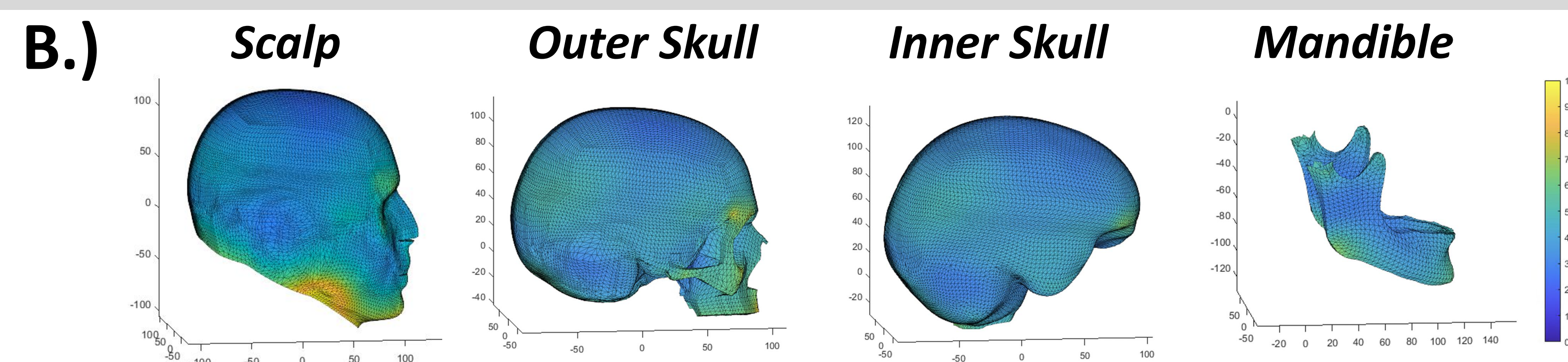


Figure 2.

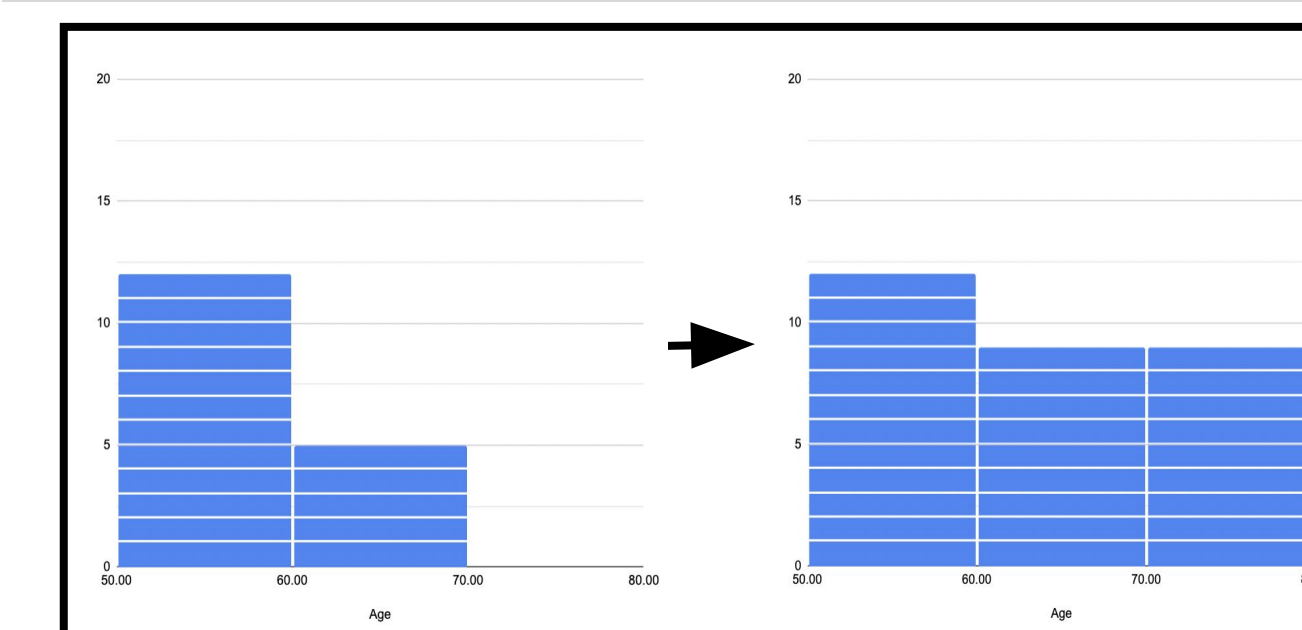
A.) Visualization and prediction results for the first three principal components and scaling factor effects on the scalp, outer skull, inner skull, and mandible geometries.

B.) Average model prediction error (mm) distributions associated with each node across the sampled subjects for the scalp, outer skull, inner skull, and mandible layers.

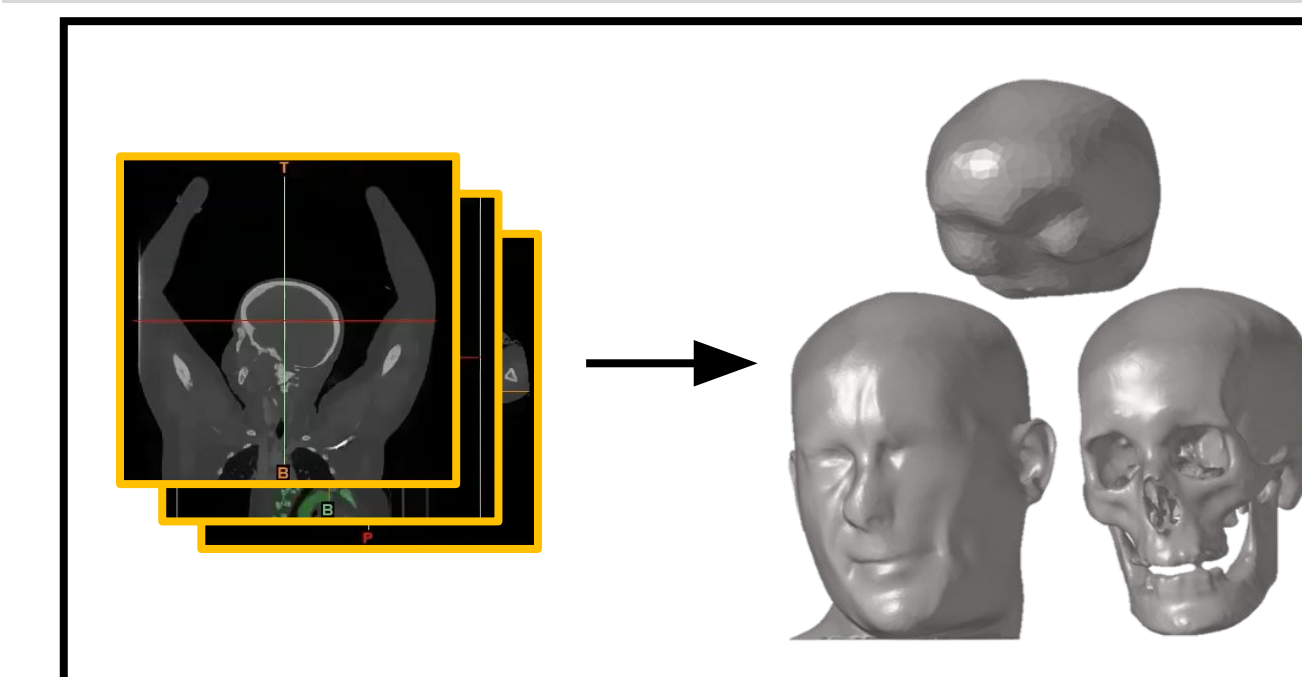


Methods

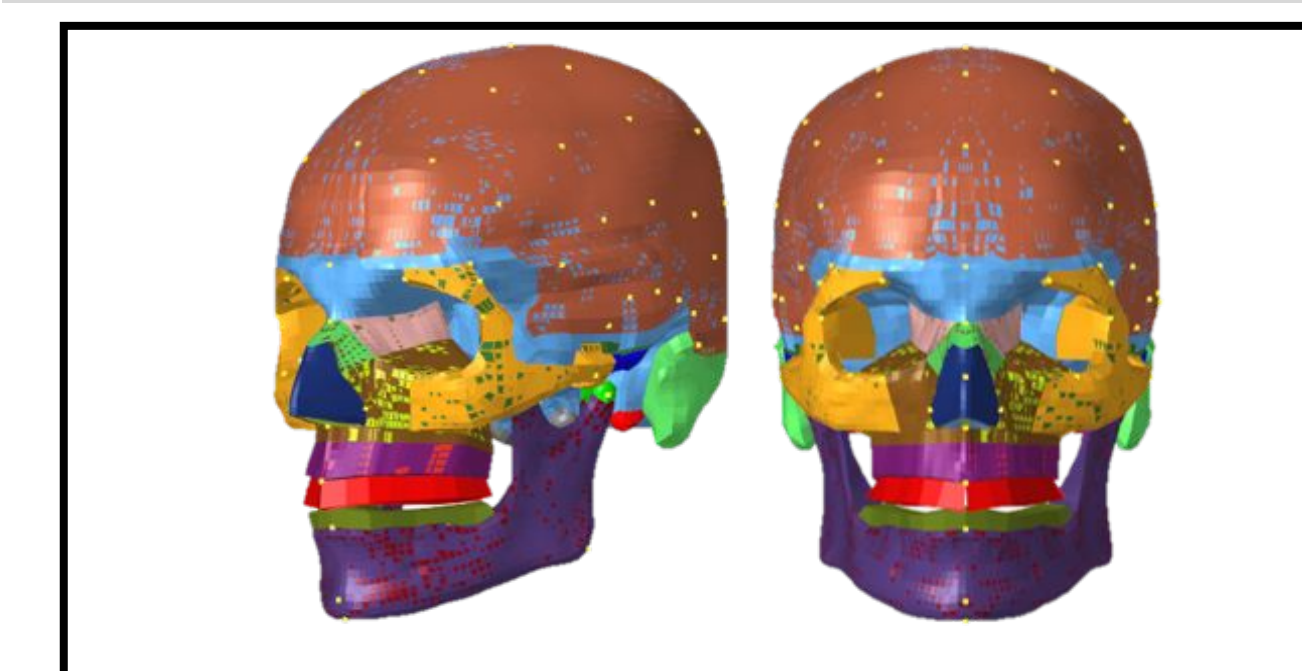
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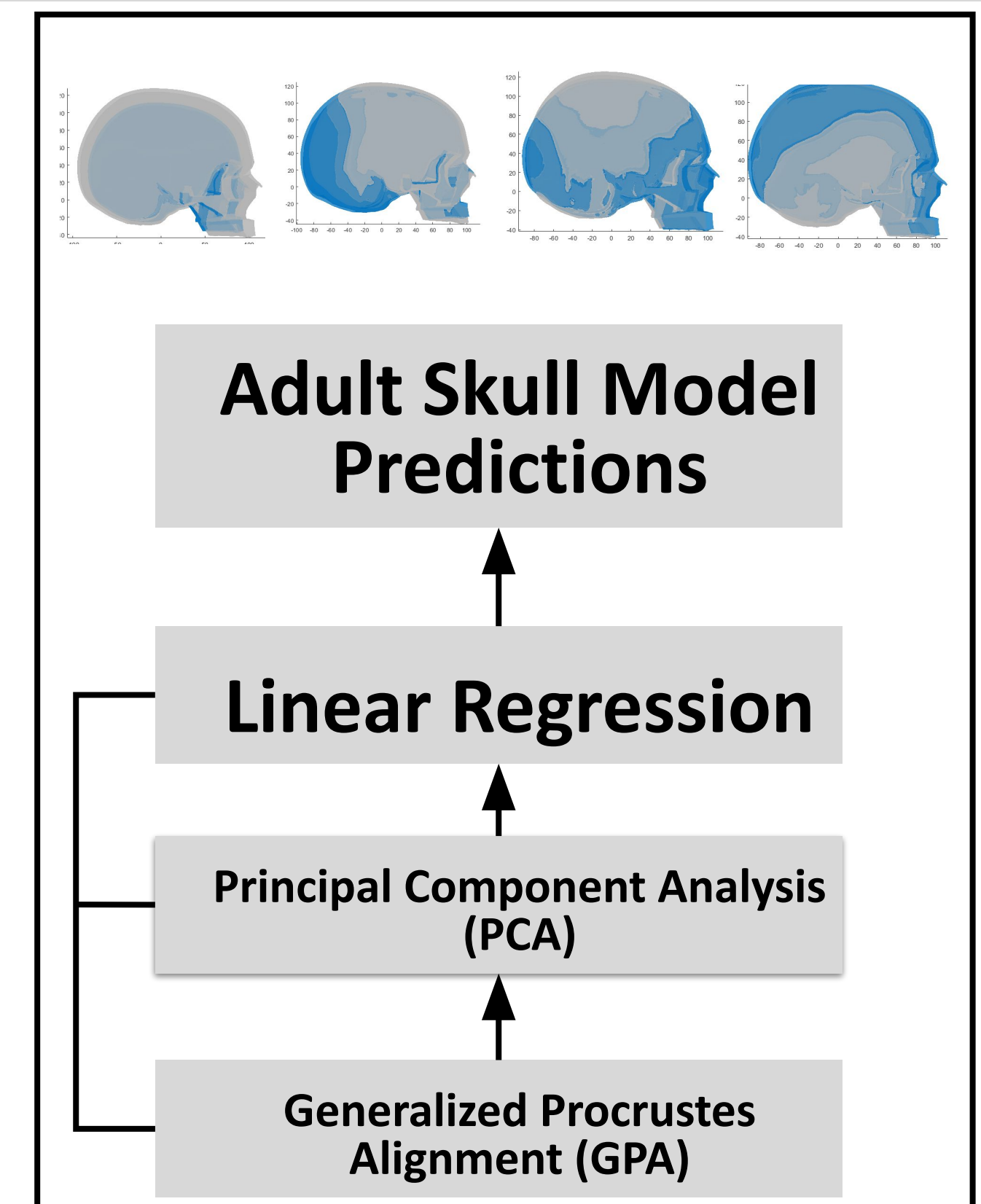
Subject Geometry Extraction



Landmarking



Statistical Analysis



Mesh-morphing and Projection

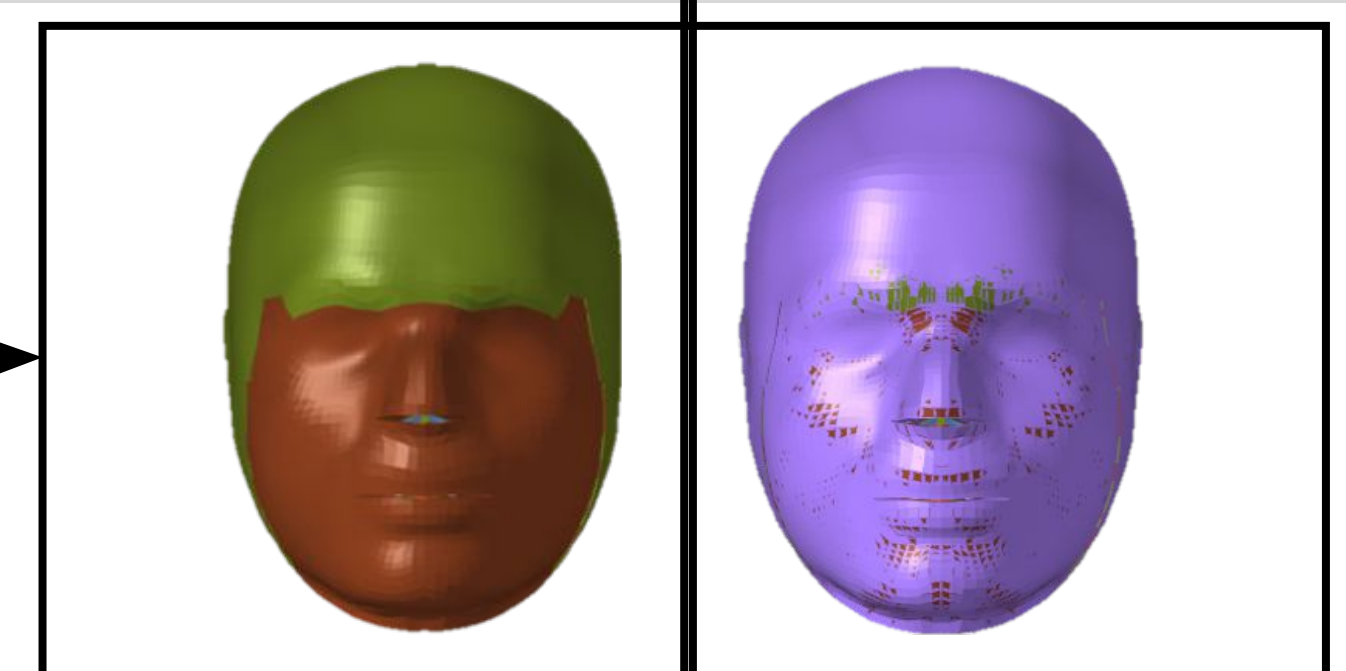


Figure 1. Methodology for generating statistical shape models.

Discussion

- Notably, the study expands the age range to 79 years.
- Key limitations of our study include: skin distortion due to the prone position in which CT scans are taken, preventing us from directly using facial skin in our model.
- Future work will be focused on applying this statistical model to baseline FE models used in risk and injury assessment, as well as making further improvements to the existing mode.

Acknowledgements

The Adult Skull team thanks Dr. Jingwen Hu and Dr. Monica Jones for their invaluable guidance. The completion of this project was facilitated by support from the University of Michigan Transportation Research Institute (UMTRI) and the Multidisciplinary Design Program (MDP).